

Total marks – 120

Attempt Questions 1 – 8

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.

Question 1 (15 marks) Use a separate piece of paper

Marks

a) Find $\int_0^{\pi} \frac{\sin \theta d\theta}{\cos^2 \theta}$

2

b) By completing the square, evaluate $\int_{-1}^0 \frac{dx}{\sqrt{1-2x-x^2}}$

3

c) (i) Find A , B and C if $\frac{16}{(x^2+4)(2-x)} = \frac{Ax+B}{(x^2+4)} + \frac{C}{(2-x)}$

2

(ii) Hence find $\int \frac{16}{(x^2+4)(2-x)} dx$

2

d) Use the substitution $x = 3\sin \theta$ to find $\int \frac{x^2}{\sqrt{9-x^2}} dx$

3

e) Using $t = \tan \frac{x}{2}$ find $\int \frac{dx}{1+\sin x}$

3

Question 2 (15 marks) Use a separate piece of paper

a) Let $z = 2 - i$ and $\omega = 1 + 3i$, find;

(i) $z\bar{\omega}$

1

(ii) $\frac{2}{z}$

1

b) Let $\alpha = \frac{4i}{-1+i\sqrt{3}}$

(i) Express α in modulus argument form

2

(ii) Express α^5 in modulus argument form

2

(iii) Hence express α^5 in the form $x + iy$

1

c) Sketch the region on the Argand diagram where the inequalities

3

$$\frac{\pi}{4} \leq \arg(z-i) \leq \frac{3\pi}{4} \quad \text{and} \quad |z-i| \leq 2$$

hold simultaneously

Question 2 (continued)

Marks

d) The point A represents the real number $z_1 = 1$. The point B represents the complex number $z_2 = (1 + \sqrt{3}) + i$. If $ABCD$ is a square in anti-clockwise rotational order, find;

(i) the complex number z_3 represented by the point D .

2

(ii) the complex number z_4 represented by the point C .

2

(iii) the complex number z_5 represented by the point of intersection of the diagonals AC and BD .

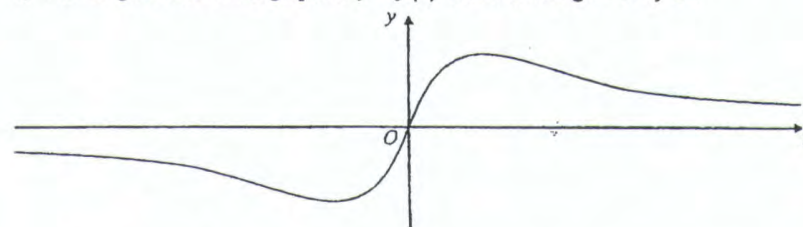
1

Question 3 (15 marks) Use a separate piece of paper

a) Sketch $y = \frac{3x^2}{4-x^2}$ showing all asymptotes.

3

b) The diagram shows the graph of $y = f(x)$ which has range $-1 \leq y \leq 1$



Draw separate one-third page sketches of the graphs of the following;

(i) $y = |f(x)|$

1

(ii) $y = f(x+1)$

1

(iii) $y = [f(x)]^2$

2

(iv) $y = \frac{1}{f(x)}$

2

(v) $y = e^{f(x)}$

2

c) It is known that $P(x) = x^4 - 4x^3 + 5x^2 + ax + b$ is divisible by $(x-2)^2$

(i) Find the values of a and b .

3

(ii) Factorise $P(x)$ into linear factors

1

Question 4 (15 marks) Use a separate piece of paper

Marks

- a) A curve has the equation $x^3 + 2xy - 4y^2 = 10$.
Find an expression for $\frac{dy}{dx}$ as a function of x and y . 2
- b) A cylindrical hole of radius 1 cm is bored through the centre of a sphere of radius 3 cm. Using the method of cylindrical shells, calculate the exact volume of the sphere that remains. 4
- c) For the conic $\frac{x^2}{25} + \frac{y^2}{16-\lambda} = 1$ find ;
(i) the values of λ that makes the conic an ellipse with the foci on the x axis. 2
(ii) the values of λ that makes the conic a rectangular hyperbola. 1
(iii) the coordinates of the foci and the equations of the directrices when $\lambda = 0$ 2
- d) How many different ways are there of seating four married couples at a circular table with men and women in alternate positions and no wife next to her husband? 4

Question 5 (15 marks) Use a separate piece of paper

- a) Suppose α, β and γ are the three roots of the polynomial equation $x^3 + x + 12 = 0$
(i) Find $\alpha^2 + \beta^2 + \gamma^2$ 2
(ii) Hence explain why only one of the roots is real 1
(iii) The real root is denoted by α . Prove that $-3 < \alpha < -2$ 1
(iv) Hence prove that the modulus of each of the other roots lies between 2 and $\sqrt{6}$ 3
- b) A woman of mass M kg jumps vertically (feet first) from a rock ledge into a river below.
When she is falling at v m/s, she encounters air resistance equal to $\frac{Mv}{10}$ Newtons.
She hits the water at a speed of V m/s.
Let x be the displacement below the rock ledge at time t seconds after jumping.
(i) Show that $\ddot{x} = g - \frac{v}{10}$, where g is the acceleration due to gravity 1
(ii) If it takes one second for her feet to hit the water, using $g = 10 \text{ m/s}^2$ show that;
$$V = 100 \left(1 - e^{-\frac{1}{10}} \right)$$

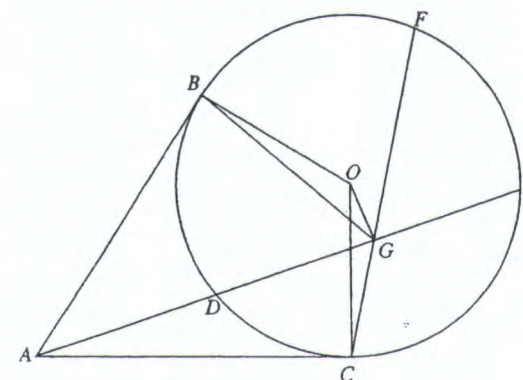
(iii) Find the height of the rock ledge above the water, to the nearest 0.1 metre 4

Question 6 (15 marks) Use a separate piece of paper

Marks

- a) By considering the series $1 + t + t^2 + t^3 + \dots + t^n$, or otherwise;
(i) sum the series $1 + 2t + 3t^2 + 4t^3 + \dots + nt^{n-1}$ 2
(ii) hence evaluate $1 + 2 \times 2 + 3 \times 2^2 + 4 \times 2^3 + \dots + 20 \times 2^{19}$ 1

b)



In the diagram, AB and CA are the tangents from A to the circle with centre O , meeting the circle at B and C .

ADE is a secant of the circle. G is the midpoint of DE . CG produced meets the circle at F .

(Note: on page 10 there is a copy of the above diagram that you can use, please detach and include with your solutions)

- (i) Show that $ABOC$ is a cyclic quadrilateral 2
(ii) Show that $AOGC$ is a cyclic quadrilateral 2
(iii) Hence prove that $BF \parallel AE$ 3

Question 6 (continued)

Marks

- c) The roof of a sports stadium has an elliptical base with a major axis of length $2a$ and minor axis of length $2b$. The two identical sloping tops are inclined at 30° to the base.

$PQRS$ represents a rectangular cross-section of thickness δx taken x units from the centre O of the ellipse.

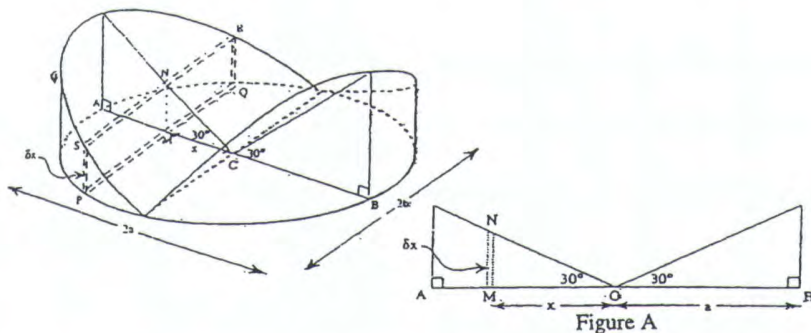


Figure A shows a side view of the stadium if sliced in half along AB

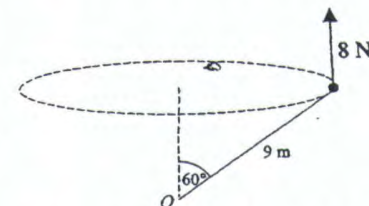
- Find the height MN of the rectangular cross-section $PQRS$ 1
- Show that the area A of the rectangle $PQRS$ is given by $A = \frac{2b}{a\sqrt{3}} x\sqrt{a^2 - x^2}$ 2
- Calculate the volume of the stadium roof. 2

24

Question 7 (15 marks) Use a separate piece of paper

Marks

a)



A toy aircraft of mass 0.5 kg is attached to one end of a string of length 9 m . The other end of the string is attached to a fixed point O and moves with constant angular velocity.

The string is taut, and makes an angle of 60° with the upward vertical at O .

In a simplified model of the motion, the aircraft is treated as a particle and the force of the air on the aircraft is taken to act vertically upwards with magnitude 8 Newtons . (Use $g = 10 \text{ m/s}^2$)

- Resolve the forces on the aircraft in the horizontal and vertical directions 2
- Find the tension in the string 1
- Find the speed of the aircraft in m/s . 2

b) $P\left(cp, \frac{c}{p}\right)$ is a point on the hyperbola $xy = c^2$

- Show that the equation of the tangent at P has the equation $x + p^2y = 2cp$ 2
- The tangent at P cuts the x axis and y axis at A and B respectively. Find the coordinates of A and B . 1
- Q is the fourth vertex of the rectangle $OAQB$. Show that the locus of Q is another rectangular hyperbola. 2

c) (i) Let $I_n = \int \tan^n \theta d\theta$, show that $I_n = \frac{1}{n-1} \tan^{n-1} \theta - I_{n-2}$ 3

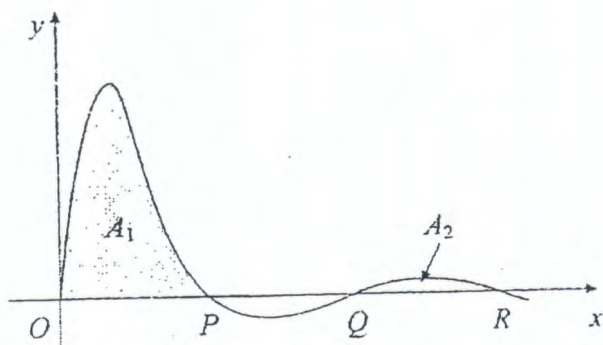
(ii) Hence find the exact value of $\int_0^{\frac{\pi}{4}} \tan^4 \theta d\theta$ 2

Question 8 (15 marks) Use a separate piece of paper

a) (i) Given that $y = x \sin x$, find $\frac{d^2 y}{dx^2}$

(ii) Prove by induction that $\frac{d^{2n} y}{dx^{2n}} = (-1)^n (x \sin x - 2n \cos x)$

b)



The diagram shows a sketch of part of the curve C with equation

$$y = e^{-x} \sin x, \quad x \geq 0$$

(i) Find the coordinates of the points P , Q and R where C cuts the x axis.

(ii) Use integration by parts to show that;

$$\int e^{-x} \sin x \, dx = -\frac{1}{2} e^{-x} (\sin x + \cos x) + c$$

(iii) The terms $A_1, A_2, \dots, A_n$ represent areas between C and the x axis for successive portions of C where y is positive. The areas represented by A_1 and A_2 are shown in the diagram.

Show that $A_n = \frac{1}{2} (e^{(1-2n)\pi} + e^{(2-2n)\pi})$

(iv) Show that $A_1 + A_2 + A_3 + \dots$ is a geometric series and that $S_\infty = \frac{e^\pi}{2(e^\pi - 1)}$

(v) Given that $\int_0^\pi e^{-x} \sin x \, dx = \frac{1}{2}$, find the exact value of

$$\int_0^\pi e^{-x} \sin x \, dx$$

Marks

1

3

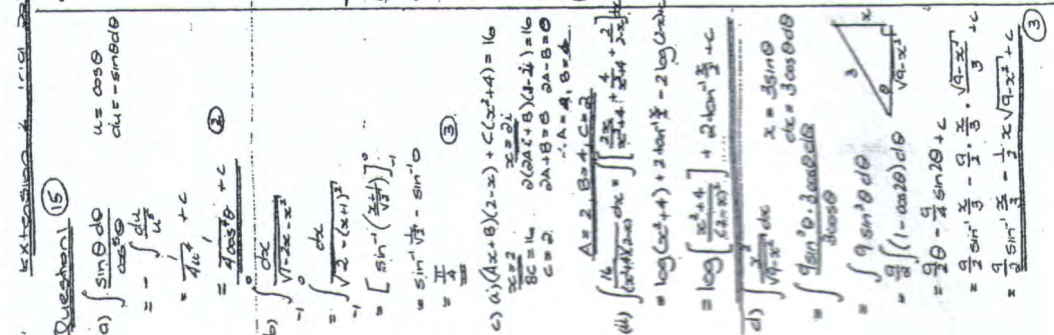
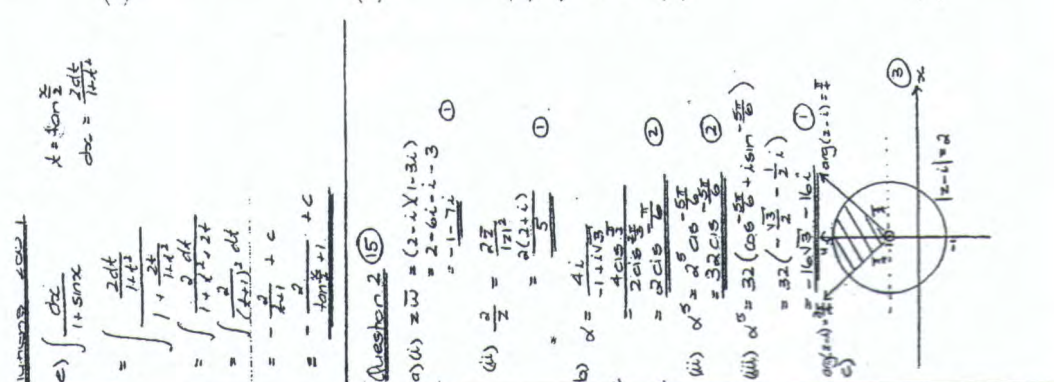
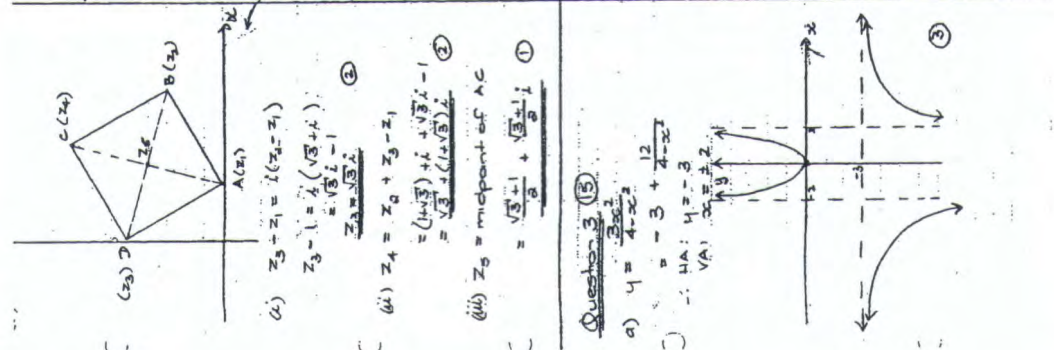
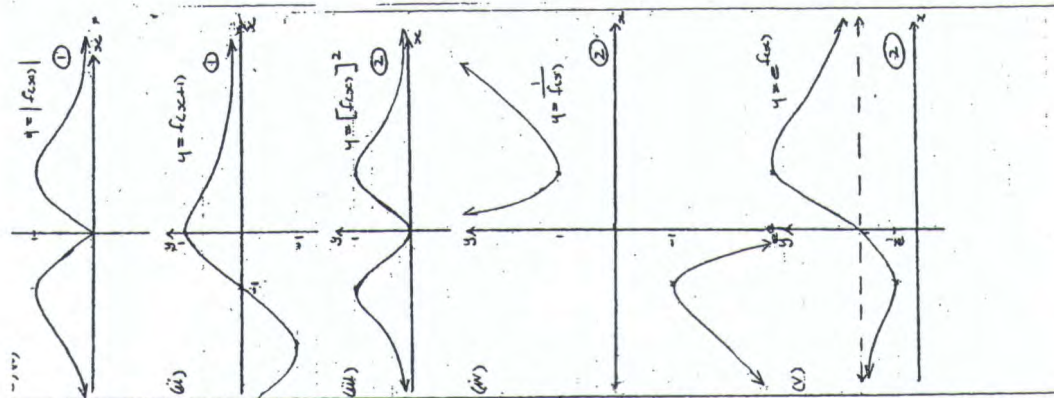
1

3

2

3

2



G(4) $P(x) = x^3 - 7x^2 + 10x + a$
 $P(x) = 4x^3 - 12x^2 + 10x + a$

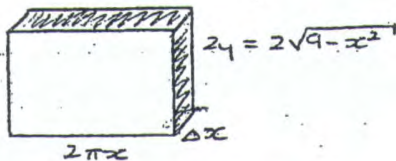
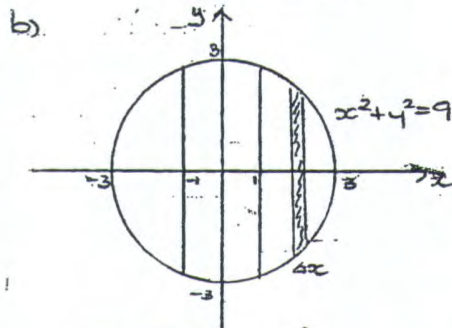
$P(2) = 0$
 $16 - 32 + 20 + 2a + b = 0$
 $2a + b = -4$

$P'(2) = 0$
 $32 - 48 + 20 + a = 0$
 $a = -4, \therefore b = 4$ (3)

(iii) $P(x) = x^4 - 4x^3 + 5x^2 - 4x + 4$
 $= (x^2 + 4x + 4)(x^2 + 1)$
 $= (x+2)^2(x+i)(x-i)$ (1)

Question 4 (15)

a) $x^3 + 2xy - 4y^2 = 10$
 $3x^2 + (2x)\frac{dy}{dx} + (y)(2) - 8y\frac{dy}{dx} = 0$
 $(2x - 6y)\frac{dy}{dx} = -3x^2 - 2y$
 $\frac{dy}{dx} = \frac{3x^2 + 2y}{6y - 2x}$ (2)



$A(x) = 4\pi x \sqrt{9-x^2}$
 $\Delta V = 4\pi x \sqrt{9-x^2} \Delta x$
 $V = \lim_{\Delta x \rightarrow 0} \sum_{x=1}^3 4\pi x \sqrt{9-x^2} \Delta x$
 $= 2\pi \int_1^3 2x \sqrt{9-x^2} dx$
 $= -2\pi \left[\frac{2}{3}(9-x^2)^{3/2} \right]_1^3$
 $= -\frac{4\pi}{3}(0 - 8\sqrt{8})$
 $= \frac{32\pi\sqrt{2}}{3}$ (4)

(i) $-16 < \lambda < 9$
 $-9 < \lambda < 16$ (2)
(ii) $16 - \lambda = -25$
 $\lambda = 41$ (1)
(iii) $\frac{x^2}{25} + \frac{y^2}{16} = 1$
 $a^2 = 25, b^2 = 16$
 $a = 5, b = 4$
 $1 - e^2 = \frac{b^2}{a^2} = \frac{16}{25}$
 $e^2 = \frac{9}{25}$
 $e = \frac{3}{5}$ (2)
 $\therefore$ foci $(\pm 3, 0)$, directrices $x = \pm \frac{25}{3}$

d) Place men first = 3!
 A Wife A has 2 spots
 B D this leaves 1 spot for
 C (M) each of the other
 wives
 Ways = $3! \times 2 \times 1 \times 1 \times 1$
 $= 12$ (4)

Question 5 (15)

a) (i) $x^3 + x + 12 = 0$
 $\sum x^2 = (\sum x)^2 - 2\sum x\beta$
 $= 0^2 - 2(1)$
 $= -2$ (2)
(ii) as $\sum x^2 < 0$ there must be an imaginary root.
 However when coefficients are real, roots appear in conjugate pairs
 $\therefore$ there are two imaginary roots
 $\therefore$ there is one real root (1)

(iii) $P(-3) = -27 - 3 + 12 = -18 < 0$
 $P(-2) = -8 - 2 + 12 = 2 > 0$
 as polynomial is continuous
 $-3 < \alpha < -2$ (1)

(iv) $\alpha\beta\gamma = -12$
 $4 < \beta\gamma < 6$
 $\beta\gamma$ are conjugates
 $z\bar{z} = |z|^2$
 $\therefore 4 < |z|^2 < 6$
 $2 < |z| < \sqrt{6}$ (3)

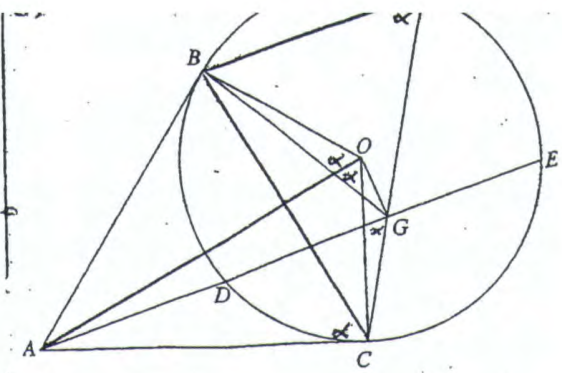
25
 $M\ddot{x} = Mg - \frac{mv}{10}$
 $\ddot{x} = g - \frac{v}{10}$ (1)
(ii) $\frac{dv}{dt} = \frac{10g - v}{10}$
 $t = -10 \int \frac{-dv}{10g - v}$
 when $t = 1, v = V$
 $1 = -10 \int_0^V \frac{-dv}{10g - v}$
 $-\frac{1}{10} = [\log(10g - v)]_0^V$
 $= \log\left(\frac{10g - v}{10g}\right)$
 $e^{-\frac{1}{10}} = \frac{10g - v}{10g}$
 $\log e^{-\frac{1}{10}} = \log\left(\frac{10g - v}{10g}\right)$
 $V = 10g(1 - e^{-\frac{1}{10}})$ (3)
 $V = 100(1 - e^{-\frac{1}{10}})$ ($g = 10$)

(iii) $v \frac{dv}{dx} = \frac{10g - v}{10}$
 $x = 10 \int \frac{v dv}{10g - v}$
 $x = -10 \int \left(1 - \frac{10g}{10g - v}\right) dv$
 $= -10 \left[v + 10 \log\left(\frac{10g - v}{10g}\right) \right]_0^V$
 $= -10 \left(V + 10 \log\left(\frac{10g - V}{10g}\right) \right)$
 $= -10 \left(100(1 - e^{-\frac{1}{10}}) + 100(-\frac{1}{10}) \right)$
 $= 4.8m$ (4)

Question 6 (15)

a) (i) $1 + t + t^2 + \dots + t^n = \frac{t^{n+1} - 1}{t - 1}$
 $t + 2t + 3t^2 + \dots + nt^n = \frac{(t-1)(n+1)t^{n+1} - (t^{n+1} - 1)}{(t-1)^2}$
 $= \frac{(n+1)t^{n+1} - nt^{n+1} - t^{n+1} + 1}{(t-1)^2}$
 $= \frac{1 - t^{n+1}}{(t-1)^2}$ (2)

(ii) $t = 2, n = 20$
 $1 + 2 \times 2 + 3 \times 2^2 + \dots + 20 \times 2^{19}$
 $= 19 \times 2^{20} - 20 \times 2^{19} + 1$
 $= 9427145$ (1)



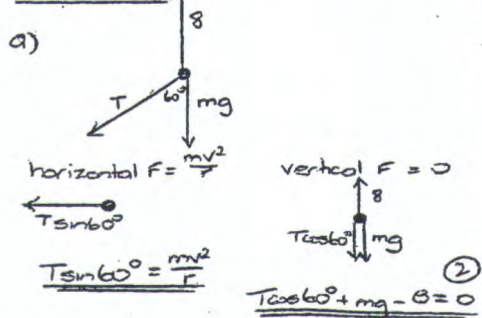
(i) $\angle OBA = 90^\circ$ (radius $\perp$ tangent)
 $\angle OCA = 90^\circ$ (radius $\perp$ tangent)
 $\therefore$ ABOC is cyclic quadrilateral (2)
 (opposite $\angle$'s supplementary)
(ii) $\angle OCA = 90^\circ$ ($\perp$ centre bisects chord)
 $\therefore \angle OCA = \angle OBA$
 AOCB is cyclic quadrilateral (2)
 ($\angle$'s in same segment =)

(iii) Let $\angle BFC = x$
 $\angle BOC = 2x$ (at centre twice at circumference on same arc)
 $\angle BCA = x$ (all in alternate segment)
 $\therefore \angle BOA = x$ ($\angle$'s in same segment = in ABOC)
 $\angle BOC = \angle AOB + \angle AOC$ (common $\angle$)
 $2x = x + \angle AOC$
 $\angle AOC = x$
 $\angle AOC = \angle ACC$ ($\angle$'s in same segment = in AOC)
 $\therefore \angle AGC = x$
 $\angle AGC = \angle BFC = x$ (3)
 $\therefore BF \parallel AE$ (corresponding $\angle$'s =)

c) $\frac{MN}{x} = \tan 30^\circ$
 $\therefore MN = \frac{x}{\sqrt{3}}$ (1)
(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 $a^2 y^2 = a^2 b^2 - b^2 x^2$
 $y^2 = \frac{b^2(a^2 - x^2)}{a^2}$
 $2y = \frac{2b\sqrt{a^2 - x^2}}{a}$
 $A(x) = \frac{x}{\sqrt{3}} \times \frac{2b\sqrt{a^2 - x^2}}{a}$
 $= \frac{2b}{a\sqrt{3}} x \sqrt{a^2 - x^2}$ (2)

$$\begin{aligned}
 \text{iii) } \Delta V &= \frac{2b}{\sqrt{3}} \int_0^a \frac{1}{\sqrt{a^2-x^2}} dx \\
 V &= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n \frac{2b}{\sqrt{3}} \cdot x \sqrt{a^2-x^2} \Delta x \\
 &= \frac{2b}{\sqrt{3}} \int_0^a 2x \sqrt{a^2-x^2} dx \\
 &= \frac{-2b}{\sqrt{3}} \left[\frac{1}{3} (a^2-x^2)^{\frac{3}{2}} \right]_0^a \\
 &= \frac{4b}{3\sqrt{3}} (a^3) \\
 &= \frac{4a^2b}{3\sqrt{3}} \text{ units}^3 \quad (2)
 \end{aligned}$$

Question 7 (15)



ii) $\frac{T}{2} = 8 - (0.5)(10)$

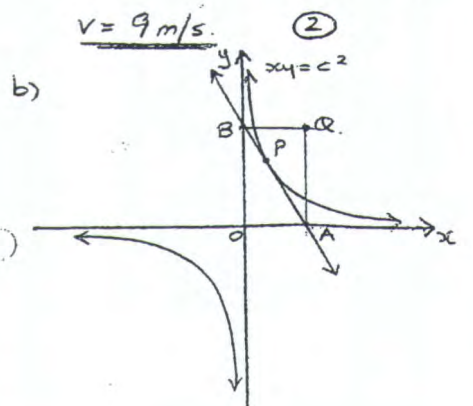
$T = 6 \text{ N}$ (1)

iii) $\frac{\sqrt{3}T}{2} = \frac{mv^2}{r}$

$\frac{\sqrt{3}(6)}{2} = \frac{(0.5)v^2}{\frac{9\sqrt{3}}{2}}$

$v^2 = 3\sqrt{3} \times \frac{9\sqrt{3}}{2} \times 2$

$v = 9 \text{ m/s}$ (2)



iv) $y = x$

$\frac{dy}{dx} = \frac{-c^2}{x^2}$

when $x = cp$, $\frac{dy}{dx} = -\frac{1}{p^2}$

$y - \frac{c}{p} = -\frac{1}{p^2} (x - cp)$

$p^2y - pc = -x + cp$

$x + p^2y = 2cp$ (2)

ii) A: $y=0$ B: $x=0$

$\therefore x = 2cp$ $p^2y = 2cp$

$y = \frac{2c}{p}$

A: $(2cp, 0)$ B: $(0, \frac{2c}{p})$ (1)

iii) Q $(2cp, \frac{2c}{p})$

$xy = 2cp \times \frac{2c}{p}$

$xy = 4c^2$

$\therefore$ locus of Q is $xy = 4c^2$ (2)

c) i) $I_n = \int \tan^n \theta d\theta$

$= \int \tan^{n-2} \theta (\sec^2 \theta - 1) d\theta$

$= \int \tan^{n-2} \theta \sec^2 \theta d\theta - \int \tan^{n-2} \theta d\theta$

$= \int u^{n-2} du - I_{n-2}$ $u = \tan \theta$ $du = \sec^2 \theta d\theta$

$= \frac{u^{n-1}}{n-1} - I_{n-2}$

$= \frac{1}{n-1} \tan^{n-1} \theta - I_{n-2}$ (3)

ii) $\int_0^{\frac{\pi}{4}} \tan^4 \theta = I_4$

$= \left[\frac{1}{3} \tan^3 \theta \right]_0^{\frac{\pi}{4}} - I_2$

$= \left[\frac{1}{3} \tan^3 \theta - \tan \theta \right]_0^{\frac{\pi}{4}} + I_0$

$= \left[\frac{1}{3} \tan^3 \theta - \tan \theta + \theta \right]_0^{\frac{\pi}{4}}$

$= \frac{1}{3} - 1 + \frac{\pi}{4}$

$= \frac{\pi}{4} - \frac{2}{3}$

$= \frac{3\pi - 8}{12}$ (2)

i) i) $y = x \sin x$

$\frac{dy}{dx} = (x)(\cos x) + (\sin x)(1)$

$= x \cos x + \sin x$

$\frac{d^2y}{dx^2} = (x)(-\sin x) + (\cos x)(1) + \cos x$

$= 2 \cos x - x \sin x$ (1)

ii) Prove true for $n=1$

LHS = $\frac{d^2y}{dx^2} = 2 \cos x - x \sin x$

RHS = $(-1)(x \sin x - 2 \cos x) = 2 \cos x - x \sin x$

$\therefore$ LHS = RHS

Hence the result is true for $n=1$

Assume the result is true for $n=k$

$\therefore \frac{d^{2k}y}{dx^{2k}} = (-1)^k (x \sin x - 2k \cos x)$

Prove the result is true for $n=k+1$

$\therefore \frac{d^{2k+2}y}{dx^{2k+2}} = (-1)^{k+1} (x \sin x - 2(k+1) \cos x)$

Proof:

$\frac{d^{2k+2}y}{dx^{2k+2}} = (-1)^k \left[(x)(\cos x) + \sin x + 2 \cos x \right]$

$= (-1)^k (x \cos x + (2k+2) \sin x)$

$\frac{d^{2k+2}y}{dx^{2k+2}} = (-1)^k \left[(x)(-\sin x) + (\cos x)(1) + (2k+2) \cos x \right]$

$= (-1)^k (-x \sin x + (2k+2) \cos x)$

$= (-1)^{k+1} (x \sin x - 2(k+1) \cos x)$

Hence the result is true for $n=k+1$ if it is also true for $n=k$

Hence the result is true for all positive integral values of n by induction. (3)

b) i) $e^{-x} \sin x = 0$

$\sin x = 0$

$x = 0, \pi, 2\pi, \dots$

$\therefore P(\pi, 0), Q(2\pi, 0), R(3\pi, 0)$ (1)

ii) $\int e^{-x} \sin x dx$

$u = e^{-x}$ $v = -\cos x$

$du = -e^{-x} dx$ $dv = \sin x dx$

$= -e^{-x} \cos x - \int e^{-x} \cos x dx$

$u = e^{-x}$ $v = \sin x$

$du = -e^{-x} dx$ $dv = \cos x dx$

$\therefore 2 \int e^{-x} \sin x dx = -e^{-x} (\sin x + \cos x) + c$

$\int e^{-x} \sin x dx = -\frac{1}{2} e^{-x} (\sin x + \cos x) + \frac{c}{2}$ (3)

iii) $A_1 = \int_0^{\pi} e^{-x} \sin x dx$

$A_2 = \int_{2\pi}^{3\pi} e^{-x} \sin x dx$

$\therefore A_n = \int_{(2n-2)\pi}^{(2n-1)\pi} e^{-x} \sin x dx$

$= -\frac{1}{2} \left[e^{-x} (\sin x + \cos x) \right]_{(2n-2)\pi}^{(2n-1)\pi}$

$= -\frac{1}{2} \left(e^{-(2n-1)\pi} - e^{-(2n-2)\pi} \right)$

$= \frac{1}{2} (e^{-(2n-2)\pi} - e^{-(2n-1)\pi})$ (2)

iv) $A_1 = \frac{1}{2} (e^{-\pi} + 1)$

$A_2 = \frac{1}{2} (e^{-3\pi} + e^{-2\pi})$

$A_3 = \frac{1}{2} (e^{-5\pi} + e^{-4\pi})$

$\therefore$ geometric series $r = e^{-2\pi}$

$S_{\infty} = \frac{\frac{1}{2} (e^{-\pi} + 1)}{1 - e^{-2\pi}}$

$= \frac{e^{-\pi} + 1}{2(e^{\pi} - 1)}$

$= \frac{e^{\pi} + 1}{2(e^{\pi} - 1)}$ (3)

A_n = areas above x axis

(v) Let $A_1 + A_2 + A_3 + \dots = X$

a_n = areas under x axis

$a_1 + a_2 + a_3 + \dots = Y$

$\therefore \int_0^{\infty} e^{-x} \sin x dx = X - Y$

$\frac{1}{2} = \frac{e^{-\pi}}{2(e^{\pi} - 1)} - Y$

$Y = \frac{e^{-\pi} - (e^{\pi} - 1)}{2(e^{\pi} - 1)}$

$= \frac{1}{2(e^{\pi} - 1)}$

$\int_0^{\infty} |e^{-x} \sin x| dx = X + Y$

$= \frac{e^{\pi}}{2(e^{\pi} - 1)} + \frac{1}{2(e^{\pi} - 1)}$

$= \frac{e^{\pi} + 1}{2(e^{\pi} - 1)}$ (2)